DP Computer Science: A3.2.5 Normal Forms

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Explain the difference between **normal forms (1NF, 2NF, 3NF)**
- Define and apply the terms **atomicity, unique identification, functional dependencies, partial-key dependencies, and non-key/transitive dependencies**
- Describe how **normalisation** addresses **data duplication, missing data, and dependency concerns**

Key Vocabulary

Term	Definition
Normalisation	The process of organising data in a database to reduce redundancy and improve integrity
1NF (First Normal Form)	Each cell contains a single value (atomic); no repeating groups
2NF (Second Normal Form)	In 1NF, and all non-key attributes are fully functionally dependent on the entire primary key
3NF (Third Normal Form)	In 2NF, and there are no transitive dependencies (non-key attributes depend only on the primary key)
Atomicity	Each attribute must contain a single, indivisible value
Functional dependency	An attribute's value is determined by another attribute (e.g., `Name` is determined by `StudentID`)
Partial-key dependency	A non-key attribute depends on only part of a composite primary key
Transitive dependency	A non-key attribute depends on another non-key attribute
Composite key	A primary key made up of two or more columns
Data redundancy	Duplicate data stored in multiple places

IB ATL Skills

Skill	Description
Thinking	Critical thinking (analysing table structures, identifying dependencies, evaluating normalisation); application (normalising tables to 3NF); problem-solving (resolving normalisation issues)
Communication	Using precise terminology; explaining rationale for normalisation decisions; presenting normalised table designs
Self-Management	Organisation (structuring data effectively); time management

2. Introduction to Normalisation

Why Normalise?

> Normalisation is the process of organising data to reduce redundancy, improve data integrity, and eliminate dependency issues.

The Problems Normalisation Solves

```text
┌─────────────────────────────────────────────────────────────────┐
│              PROBLEMS NORMALISATION SOLVES                     │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │              DATA DUPLICATION                          │  │
│    │                                                         │  │
│    │   Same data stored in multiple places                 │  │
│    │   Wastes storage; update anomalies                    │  │
│    └─────────────────────────────────────────────────────────┘  │
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │              MISSING DATA                              │  │
│    │                                                         │  │
│    │   Incomplete records due to poor design                │  │
│    │   NULL values create confusion                        │  │
│    └─────────────────────────────────────────────────────────┘  │
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │              DEPENDENCY ISSUES                         │  │
│    │                                                         │  │
│    │   Incorrect dependencies lead to data inconsistency    │  │
│    │   Update, insertion, and deletion anomalies           │  │
│    └─────────────────────────────────────────────────────────┘  │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

The Normalisation Mantra

> *"The key, the whole key, and nothing but the key."*

Normal Form	Mantra	Meaning
1NF	"The key"	Every table has a primary key; each cell has one value
2NF	"The whole key"	No partial dependencies on a composite key
3NF	"Nothing but the key"	No transitive dependencies (non-key to non-key)

3. First Normal Form (1NF)

Rules for 1NF

Rule	Description
Atomicity	Each cell contains a single, indivisible value

Rule	Description
No Repeating Groups	No columns that contain multiple values (e.g., Phone1, Phone2)
Primary Key	Each row must be uniquely identifiable

Non-1NF Example

```text
┌─────────────────────────────────────────────────────────────────┐
│           NON-1NF TABLE                                        │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │             ORDERS                                     │  │
│    ├─────────────────────────────────────────────────────────┤  │
│    │  OrderID │ Customer  │ Products                        │  │
│    ├──────────┼───────────┼─────────────────────────────────┤  │
│    │  101     │ Alice     │ Laptop, Mouse, Keyboard        │  │
│    │  102     │ Bob       │ Phone, Charger                 │  │
│    │  103     │ Charlie   │ Monitor                        │  │
│    └──────────┴───────────┴─────────────────────────────────┘  │
│                                                                 │
│    ❌ Products column contains multiple values (not atomic)   │
│    ❌ Repeating groups present                                 │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

After Normalisation to 1NF

```text
┌─────────────────────────────────────────────────────────────────┐
│           AFTER 1NF                                            │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │             ORDER_ITEMS                                │  │
│    ├─────────────────────────────────────────────────────────┤  │
│    │  OrderID (PK) │ Customer │ Product     │               │  │
│    ├───────────────┼──────────┼─────────────┤              │  │
│    │  101          │ Alice    │ Laptop      │              │  │
│    │  101          │ Alice    │ Mouse       │              │  │
│    │  101          │ Alice    │ Keyboard    │              │  │
│    │  102          │ Bob      │ Phone       │              │  │
│    │  102          │ Bob      │ Charger     │              │  │
│    │  103          │ Charlie  │ Monitor     │              │  │
│    └───────────────┴──────────┴─────────────┘              │  │
│                                                                 │
│    ✅ Each cell has one value (atomic)                         │
│    ✅ No repeating groups                                      │
│    ✅ Primary key: (OrderID, Product) — composite key         │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

---

4. Second Normal Form (2NF)

Rules for 2NF

Rule	Description
Must be in 1NF	Already normalised to 1NF
No Partial Dependencies	All non-key attributes must depend on the **entire** primary key (not just part of it)

1NF Table with Partial Dependency

```text
┌─────────────────────────────────────────────────────────────────┐
│              1NF TABLE WITH PARTIAL DEPENDENCY                  │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │              ORDER_ITEMS                               │  │
│    │      Composite Key: (OrderID, Product)                 │  │
│    ├─────────────────────────────────────────────────────────┤  │
│    │  OrderID (PK) │ Product (PK) │ Customer │ Price      │  │
│    ├──────────────┼──────────────┼──────────┼───────────┤  │
│    │  101         │ Laptop       │ Alice    │ 1200.00  │  │
│    │  101         │ Mouse        │ Alice    │ 25.00    │  │
│    │  101         │ Keyboard     │ Alice    │ 80.00    │  │
│    │  102         │ Phone        │ Bob      │ 600.00   │  │
│    │  103         │ Monitor      │ Charlie  │ 300.00   │  │
│    └──────────────┴──────────────┴──────────┴───────────┘  │
│                                                                 │
│    ❌ Customer depends only on OrderID (partial dependency)  │
│    ❌ Price depends only on Product (partial dependency)     │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

Normalising to 2NF

```text
┌─────────────────────────────────────────────────────────────────┐
│              AFTER 2NF                                          │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │              ORDERS                                    │  │
│    │           Primary Key: OrderID                          │  │
│    ├─────────────────────────────────────────────────────────┤  │
│    │  OrderID (PK) │ Customer                              │  │
│    ├──────────────┼────────────────────────────────────────┤  │
│    │  101         │ Alice                                  │  │
│    │  102         │ Bob                                    │  │
│    │  103         │ Charlie                                │  │
│    └──────────────┴────────────────────────────────────────┘  │
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │              PRODUCTS                                  │  │
│    │           Primary Key: Product                          │  │
│    ├─────────────────────────────────────────────────────────┤  │
│    │  Product (PK) │ Price                                  │  │
│    ├──────────────┼────────────────────────────────────────┤  │
│    │  Laptop       │ 1200.00                                │  │
│    │  Mouse        │ 25.00                                  │  │
```

```
│    │  Keyboard     │ 80.00                              │  │
│    │  Phone        │ 600.00                             │  │
│    │  Monitor      │ 300.00                             │  │
│    └──────────────┴─────────────────────────────────────┘  │
│                                                            │
│    ┌─────────────────────────────────────────────────────┐ │
│    │             ORDER_ITEMS                           │  │
│    │     Composite Key: (OrderID, Product)             │  │
│    ├─────────────────────────────────────────────────────┤ │
│    │  OrderID (PK)(FK) │ Product (PK)(FK)             │  │
│    ├──────────────────┼──────────────────────────────────┤  │
│    │  101              │ Laptop                       │  │
│    │  101              │ Mouse                        │  │
│    │  101              │ Keyboard                     │  │
│    │  102              │ Phone                       │  │
│    │  103              │ Monitor                     │  │
│    └──────────────────┴──────────────────────────────────┘  │
│                                                            │
│    ✅ All non-key attributes depend on the entire primary key │
│                                                            │
└────────────────────────────────────────────────────────────┘
```

---

5. Third Normal Form (3NF)

Rules for 3NF

Rule	Description
Must be in 2NF	Already normalised to 2NF
No Transitive Dependencies	Non-key attributes must depend only on the primary key (not on other non-key attributes)

2NF Table with Transitive Dependency

```text
┌────────────────────────────────────────────────────────────┐
│              2NF TABLE WITH TRANSITIVE DEPENDENCY         │
├────────────────────────────────────────────────────────────┤
│                                                            │
│    ┌─────────────────────────────────────────────────────┐ │
│    │             ORDERS                                │  │
│    │          Primary Key: OrderID                      │  │
│    ├─────────────────────────────────────────────────────┤ │
│    │  OrderID (PK) │ CustomerID │ CustomerName │ City    │  │
│    ├──────────────┼────────────┼─────────────┼──────────┤  │
│    │  101          │ 1          │ Alice        │ London  │  │
│    │  102          │ 2          │ Bob          │ Paris   │  │
│    │  103          │ 1          │ Alice        │ London  │  │
│    └──────────────┴────────────┴─────────────┴──────────┘  │
│                                                            │
│    ❌ Transitive Dependency: City depends on CustomerID,   │
│       not OrderID                                        │
│    ❌ CustomerName depends on CustomerID, not OrderID      │
│                                                            │
└────────────────────────────────────────────────────────────┘
```

Normalising to 3NF

```text
┌─────────────────────────────────────────────────────────────────┐
│           AFTER 3NF                                           │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │           ORDERS                                      │  │
│    │        Primary Key: OrderID                            │  │
│    ├─────────────────────────────────────────────────────────┤  │
│    │  OrderID (PK) │ CustomerID (FK)                       │  │
│    ├──────────────┼───────────────────────────────────────┤  │
│    │  101         │ 1                                      │  │
│    │  102         │ 2                                      │  │
│    │  103         │ 1                                      │  │
│    └──────────────┴───────────────────────────────────────┘  │
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │           CUSTOMERS                                   │  │
│    │        Primary Key: CustomerID                         │  │
│    ├─────────────────────────────────────────────────────────┤  │
│    │  CustomerID (PK) │ CustomerName │ City                │  │
│    ├──────────────────┼──────────────┼──────────────────────┤  │
│    │  1               │ Alice        │ London              │  │
│    │  2               │ Bob          │ Paris               │  │
│    └──────────────────┴──────────────┴──────────────────────┘  │
│                                                                 │
│    ✅ No transitive dependencies                              │
│    ✅ Each non-key attribute depends only on the primary key  │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

---

6. Normal Forms Summary Table

Normal Form	Requirement	Key Focus	Mantra
1NF	Atomic values; no repeating groups	Atomicity	"The key"
2NF	In 1NF; no partial dependencies	Whole key	"The whole key"
3NF	In 2NF; no transitive dependencies	Non-key dependencies	"Nothing but the key"

Visual Summary

```text
┌─────────────────────────────────────────────────────────────────┐
│           NORMAL FORMS FLOWCHART                              │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│                      START                                      │
│                        │                                        │
│                        ▼                                        │
│              ┌─────────────────────┐                           │
│              │  Is each cell      │                           │
│              │  atomic? No        │                          │
│              │  repeating groups?│                          │
│              └─────────────────────┘                           │
│                        │                                        │
```

```
│                    Yes                                   │
│                     ▼                                    │
│              ┌──────────────┐                            │
│              │    1NF       │                            │
│              │  (Normalised to 1NF)│                     │
│              └──────────────┘                            │
│                     │                                    │
│                     ▼                                    │
│              ┌──────────────┐                            │
│              │  Any partial │                            │
│              │  dependencies│                            │
│              │  on composite│                            │
│              │  key?        │                            │
│              └──────────────┘                            │
│                     │                                    │
│                    No                                    │
│                     ▼                                    │
│              ┌──────────────┐                            │
│              │    2NF       │                            │
│              │  (Normalised to 2NF)│                     │
│              └──────────────┘                            │
│                     │                                    │
│                     ▼                                    │
│              ┌──────────────┐                            │
│              │  Any transitive │                         │
│              │  dependencies?│                           │
│              └──────────────┘                            │
│                     │                                    │
│                    No                                    │
│                     ▼                                    │
│              ┌──────────────┐                            │
│              │    3NF       │                            │
│              │  (Normalised to 3NF)│                     │
│              └──────────────┘                            │
│                                                          │
└──────────────────────────────────────────────────────────┘
```

---

7. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	5 min	"Why is organising data important?"; recap tables; introduce normalisation
1NF	15 min	Atomicity; no repeating groups; transform non-1NF table to 1NF
2NF	20 min	Functional dependencies; partial-key dependencies; normalise to 2NF
3NF	15 min	Transitive dependencies; normalise to 3NF
Wrap-up	5 min	Summary; Q&A

Normalisation Practice Table

OrderID	Customer	Product	Quantity	Price	CustomerCity
101	Alice	Laptop	1	1200	London
101	Alice	Mouse	2	25	London
102	Bob	Phone	1	600	Paris
103	Charlie	Monitor	1	300	New York

Challenge: Normalise this table to 3NF.

Solution:

1NF	2NF	3NF
Already in 1NF (no repeating groups, atomic values)	Split into: Orders (OrderID, Customer), OrderItems (OrderID, Product, Quantity), Products (Product, Price)	Split further: Orders (OrderID, CustomerID), OrderItems (OrderID, Product, Quantity), Products (Product, Price), Customers (CustomerID, CustomerName, City)

---

8. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Worksheet	Evaluate ability to normalise tables to 3NF
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the difference between 1NF and 2NF?
2. What is a partial dependency?
3. What is a transitive dependency?
4. Why is normalisation important?

---

9. Differentiation

Level	Strategy
Support	Provide step-by-step examples; use visual aids
Challenge	Normalise complex multi-table databases

---

10. Resources

- **Presentation** (Slide Deck)
- **eBook** (A3.2.5 with examples)
- **Worksheet 1:** Normal forms practice
- **Worksheet 2:** Step-by-step normalisation process

- **Worksheet Answers**
- **Quizlet** (A3.2.5 vocabulary flashcards)

---

11. Summary: Key Takeaways

Normal Form	Key Rule	Key Problem Solved
1NF	Atomic values; no repeating groups	Multi-valued attributes
2NF	No partial dependencies (on composite keys)	Data duplication; update anomalies
3NF	No transitive dependencies	Update, insertion, deletion anomalies

```text
┌─────────────────────────────────────────────────────────────┐
│                    KEY REMINDERS                            │
├─────────────────────────────────────────────────────────────┤
│                                                             │
│    ✅ 1NF: One cell, one value                            │
│    ✅ 2NF: The whole key                                  │
│    ✅ 3NF: Nothing but the key                           │
│    ✅ Partial dependency = depends on part of composite key │
│    ✅ Transitive dependency = non-key → non-key          │
│    ✅ Normalisation reduces redundancy and improves integrity│
│                                                             │
└─────────────────────────────────────────────────────────────┘
```

> *"Normalisation is the process of organising data to reduce redundancy and dependency issues—'the key, the whole key, and nothing but the key.'"*